When Walks Become a Spectrum

A machine-checked proof of Godsil's moment theorem $p_k = \text{treeLikeWalkCount}$ in Lean 4

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Abstract

This is the fourth piece of a series that builds, in Lean 4 / Mathlib, the two sides of the matching/Ihara trace formula. In *Walks that Forget the Cycles* [19] I proved the *combinatorial* half of Godsil's moment theorem: the closed *tree-like* walks of a graph G based at v are in length-preserving bijection with the closed walks of Godsil's path tree $T(G, v)$ based at its root, so that the tree-like walk count is a sum of adjacency-matrix powers, $\text{treeLikeWalkCount}(G, k) = \sum_v \left[A(T(G, v))^k\right]_{\text{root}}$. That paper was explicit about its boundary: the *spectral* half—turning those matrix powers into the power sums $p_k = \sum_i \theta_i^k$ of the roots of the matching polynomial $\mu(G)$—was *mapped, not built.* Here I build it, and close the theorem:

$$p_k(G) \;=\; \sum_i \theta_i^k \;=\; \text{treeLikeWalkCount}(G, k), \qquad \text{machine-checked, } \texttt{sorry}\text{-free.}$$

The proof is a single power-series identity. On the trace side, each path tree's root–root resolvent $\sum_k [A(T)^k]_{\text{root}} X^k$ is, by a principal-minor determinant collapse, the ratio of two reversed characteristic polynomials; Part II's divisibility identity $\mu(G)\mu(T - \text{root}) = \mu(G - v)\mu(T)$ rewrites it as a ratio of reversed *matching* polynomials; summing over v and applying the vertex-deletion law $\sum_v \mu(G - v) = X\mu'(G)$ gives $\text{mk}(\text{treeLikeWalkCount}) \cdot \overline{\mu} = \overline{X\mu'}$. On the matching side, the generating function of the root power sums equals, after a geometric-series cancellation against the reversed factor product, the *same* object $\overline{X\mu'}$. The two coincide; cancelling the unit $\overline{\mu}$ and reading the coefficient of X^k is the theorem. I record one honest surprise: the completion needs *no* univariate Newton identity—the route the prequel predicted—only a reflection-at-fixed-degree bookkeeping of reversed polynomials. The headline depends only on the three standard axioms; I am not aware of a prior formalization of Godsil's moment theorem in any proof assistant. With this and the *Bass* companion, both sides of the finite trace formula now stand, `sorry`-free, in one library.

Reader's map. If you read only one thing, read Theorem 1 and the shape of its proof: *both* generating functions, the walk-counting one and the eigenvalue-power-sum one, are forced to equal one and the same reversed polynomial $\overline{X\mu'(G)}$ (Figure 2). Everything else is the careful version of that sentence—a per-vertex identity that turns the trace side into reversed matching polynomials (Section 3), a geometric-series cancellation that turns the matching side into the same (Section 4), and a coefficient-extraction once a unit is cancelled (Section 5). The honest surprise—no Newton—is Section 7; the limits and methodology, Section 8; the next questions, Section 9.

*Formalized with AI assistance (Claude, Anthropic); the mathematics and all claims are the author's responsibility. The methodology is discussed in Section 8.

1 Introduction: a count, and the spectrum it hides

A graph's matching polynomial is not the characteristic polynomial of its adjacency matrix—so where do the power sums of its roots live, and why are they a count of walks? The prequel answered the first clause with a tree and left the second as a promissory note; this paper redeems it. The route has a long history—four centuries—and a small surprise waiting at its last step (Section 7).

A thread, in five beads. In 1629 Albert Girard wrote down what Newton would systematize a generation later: the power sums $p_k = \sum_i \theta_i^k$ of a polynomial's roots are fixed by its coefficients, and conversely, through a recursion now bearing Newton's name [7, 8]. That recursion is the canonical bridge between a polynomial's roots and its data, and it will be the bead this paper, at its very end, declines to use. In 1972 Heilmann and Lieb produced a polynomial that seemed to have no right to real roots at all—the matching polynomial of a graph, born in the statistical mechanics of monomer–dimer systems—and proved its roots real anyway [1], even though it is not the characteristic polynomial of the graph's adjacency matrix unless the graph is a forest. In 1981 Godsil resolved the paradox with a single device, the *path tree*, and drew from it *two* conclusions: that the matching polynomial is real-rooted (it divides a forest's, and forests have symmetric spectra), and that its root power sums *count walks* [4, 6]. The fourth bead is the *universal cover*: the path tree is its finite truncation, the cover's spectral edge $2\sqrt{\Delta - 1}$ is Godsil's bound on the matching roots and the Ramanujan threshold [9, 11, 15], and the walk count is a finite shadow of a trace formula on the cover. The fifth bead is this series: Part II carried Godsil's first conclusion into Lean [17], Part III carried the combinatorial half of the second [19], and this paper carries the rest—and, threading the last bead, comes back unexpectedly to the first (Section 7).

The polynomial off the adjacency spectrum. A finite graph G on n vertices carries a matching polynomial $\mu(G, x) = \sum_k (-1)^k m_k(G) x^{n-2k}$, with $m_k(G)$ the number of k-edge matchings [2, 3]. Heilmann and Lieb proved its roots $\theta_1, \ldots, \theta_n$ real [1]; but $\mu(G)$ equals the characteristic polynomial of the adjacency matrix $A(G)$ only when G is a forest.[1] Their power sums $p_k(G) = \sum_i \theta_i^k$ are therefore the moments of a *ghost* spectrum—real numbers behaving like eigenvalues, yet not those of the graph's own adjacency matrix. Godsil's moment theorem says where they come from: they *count walks*.

One tree, two theorems. Godsil's device [4, 5, 6] is the path tree $T(G, v)$, whose vertices are the simple paths of G from v and whose edges are single-edge extensions. It is a forest—a finite truncation of the universal cover unrolled from v—and Part II of this series [17] used it to machine-check Heilmann–Lieb: $\mu(G)$ divides $\mu(T(G, v))$, and a forest's matching polynomial *is* its adjacency characteristic polynomial, hence real-rooted. Part III [19] then taught the same tree to count: a closed walk of G is *tree-like* when at every step it advances to a fresh vertex or retreats to its parent, never knowingly closing a cycle, and the tree-like closed walks of G at v are exactly the closed walks of $T(G, v)$ at its root, projected down by recording the current endpoint. The consequence,

$$\boxed{\text{treeLikeWalkCount}(G, k) \ = \ \sum_v \left[A(T(G, v))^k\right]_{\text{root}},}$$

[1] Every monic polynomial is, trivially, the characteristic polynomial of its companion matrix, and—being real-rooted—of $\operatorname{diag}(\theta_1, \ldots, \theta_n)$. The point is that $\mu(G)$ is not the characteristic polynomial of the matrix built *canonically from the graph*, its adjacency matrix, unless G is a forest. We thank Scott Hotton for prompting this clarification.

is Part III's headline (`treeLikeWalkCount_eq_sum_pathTree_adjMatrix_pow`). But Part III was careful to mark its boundary: the equality of this count with the moments p_k—the *spectral* half—was mapped in its closing section, not formalized. This paper builds it. The statement being built is Godsil's, in its classical form:

Godsil's moment theorem (1981, classical form) [ ]

Let G be a finite graph on n vertices, with matching polynomial $\mu(G, x)$ of real roots $\theta_1, \dots, \theta_n$. For every $k \geq 0$,

$$p_k(G) \;=\; \sum_{i=1}^{n} \theta_i^k \;=\; \#\{\text{closed tree-like walks of length } k \text{ in } G\}.$$

Theorem 1 below is the machine-checked form of this statement.

What "building it" means. The two halves are two generating functions: $\mathrm{mk}(k \mapsto \mathrm{treeLikeWalkCount}(G, k))$ on the trace side and $\mathrm{mk}(k \mapsto p_k(G))$ on the matching side, both elements of a formal power-series ring. The theorem is that they are equal; below the girth their coefficients already agree (every short walk is tree-like), but the full statement is a single identity in $\mathbb{R}[\![X]\!]$ pushed to $\mathbb{C}[\![X]\!]$ to make room for the complex roots. The strategy is to show that each side, multiplied by the *reversed* matching polynomial $\overline{\mu(G)}$, equals one and the same reversed polynomial $\overline{X\mu'(G)}$; since $\overline{\mu(G)}$ is a unit (its constant term is the leading coefficient 1 of the monic μ), the two generating functions coincide, and the coefficient of X^k is the theorem.

What is new, and what is not. The contribution divides cleanly. The *mathematics* is Godsil's, four decades old; nothing in the classical statement above is new. What is new is, first, its *complete machine certification*—the first I am aware of in any proof assistant—and, second, a handful of *general lemmas* that the certification forced into existence and that did not previously exist in Mathlib (Table 1 traces the inputs).

Why this was not routine to formalize. A matching polynomial is not the characteristic polynomial of any single matrix, so none of the matrix machinery applies to it directly; the proof must route through Godsil's path tree and back, and every crossing is a coercion between a polynomial, its *reverse*, and a formal power series—bookkeeping Mathlib does not package. Three specific gaps had to be filled. (i) Mathlib's cofactor identity $\operatorname{adj}(M)_{ii} = \det(M{\restriction}_{j\neq i})$ existed only for the index type `Fin n`; the path-tree indices are an arbitrary finite type, so a basis-free, block-triangular proof was needed. (ii) There was no lemma that a sum of equal-degree polynomials reverses termwise ($\mathrm{reflect}_N$ is additive, rev is not), the fact that lets the per-vertex pieces be summed. (iii) The reverse of a split product, $\mathrm{rev} \prod(X - \theta) = \prod(1 - \theta X)$, and the geometric-series cancellation that replaces Newton's recursion, were likewise absent. None is deep; each is general, and all are kept graph-free for upstreaming. The headline, `matchingPowerSum_eq_treeLikeWalkCount`, depends only on `propext`, `Classical.choice`, and `Quot.sound`; there is no `sorry`, as `#print axioms` will confirm.

2 The two generating functions

What are the two objects, and in which ring do they meet? Keeping them honestly apart matters, because the whole content is that they coincide. The trace side is a count of natural numbers; the

Table 1: What this paper rests on. The two preceding parts supply the real-rootedness and the combinatorial bijection; this paper supplies the spectral completion that joins them into the moment theorem.

Ingredient	Statement	Source
Heilmann–Lieb / real-rootedness	$\mu(G) \mid \mu(T(G,v))$, forests real-rooted	Part II [17]
Tree-like walk bijection	$\mathrm{tlwc}(G,k) = \sum_v [A(T(G,v))^k]_{\mathrm{root}}$	Part III [19]
Divisibility identity	$\mu(G)\mu(T-r) = \mu(G-v)\mu(T)$	Part II [17]
Spectral completion	$p_k = \mathrm{tlwc}(G,k)$ **(the moment theorem)**	**this paper**

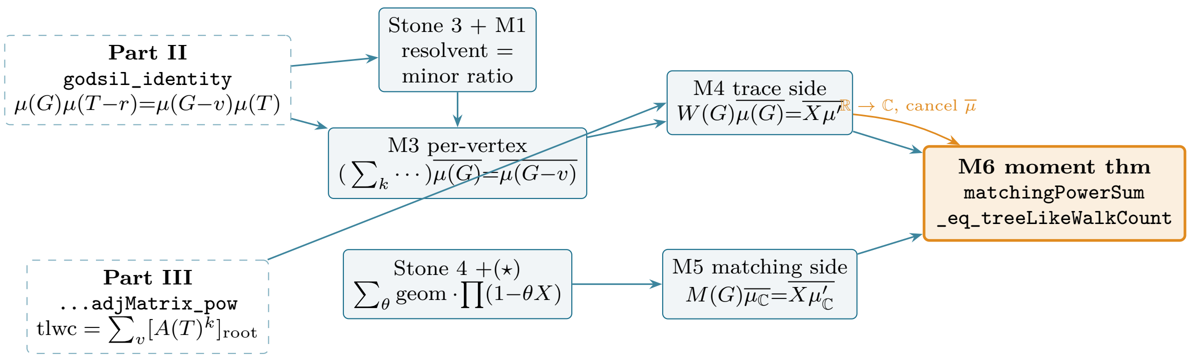

Figure 1: Formal-dependency roadmap. Two external inputs (Parts II and III, dashed) feed a trace column and a matching column; each terminates at the same reflected derivative $\overline{X\mu'}$ (M4 over $\mathbb{R}$, M5 over $\mathbb{C}$); the base change $\mathbb{R} \to \mathbb{C}$ and cancellation of the unit $\overline{\mu}$ close the moment theorem (M6, amber). Every box is a `sorry`-free Lean result; cf. Table 3.

matching side lives over $\mathbb{C}$, because the roots do.

The trace side is the formal power series whose k-th coefficient is the tree-like walk count,

$$W(G) \;:=\; \mathrm{mk}\big(k \mapsto (\mathrm{treeLikeWalkCount}(G,k) : \mathbb{R})\big) \;\in\; \mathbb{R}[\![X]\!].$$

By Part III's spectral form it is a sum of root–root resolvents of the path trees; we will keep it over $\mathbb{R}$ as long as possible and cross to $\mathbb{C}$ only at the end. The matching side is

$$M(G) \;:=\; \mathrm{mk}\big(k \mapsto p_k(G)\big) \;\in\; \mathbb{C}[\![X]\!], \qquad p_k(G) = \sum_{\theta \in \mathrm{roots}(\mu_{\mathbb{C}})} \theta^k,$$

where $\mu_{\mathbb{C}} := \mu(G)$ mapped along $\mathbb{R} \hookrightarrow \mathbb{C}$, so that its multiset of roots has exactly $n = |V(G)|$ elements (`matchingPowerSum`, `matchingPowerSum_genfun`). The proof equates the images of $W(G)$ and $M(G)$ in $\mathbb{C}[\![X]\!]$.

Throughout, $\overline{p}$ denotes the coercion of the reversed polynomial $\mathrm{rev}(p) = \mathrm{reflect}_{\deg p}(p)$ into a power-series ring, and $\mathrm{reflect}_N$ is the degree-N reflection $x^j \mapsto x^{N-j}$. The single algebraic fact we lean on repeatedly is that $\mathrm{reflect}_N$ is additive in p at a *fixed* length N, even though rev (which reflects at each polynomial's own degree) is not.

3 The trace side becomes a reversed matching polynomial

Part III left the trace side as a sum of matrix powers of trees. How does a sum over path trees, each a different forest, become a statement about G's own matching polynomial? Through three moves: a determinant collapse that identifies the resolvent's numerator, Part II's divisibility identity that trades the tree for the graph, and a reflection that sums the pieces. The walk generating function of a graph is itself a classical object [12]; what is new is the machine-checked path from it to the matching polynomial.

The per-tree resolvent. For any matrix M over a commutative ring and index i, the diagonal of the resolvent generates the diagonal matrix powers,

$$\Big(\sum_{k\geq 0}[M^k]_{ii}X^k\Big)\cdot\overline{\mathrm{charpolyRev}(M)} \;=\; \overline{\mathrm{charpolyRev}(M\!\restriction_{j\neq i})},$$

the single-entry case of Cramer's rule, where $\mathrm{charpolyRev}(M) = \det(1 - XM)$ is the reversed characteristic polynomial (`resolventGenfun_diag_mul_coe_charpolyRev`). Its proof needed one general lemma absent from Mathlib: the principal-minor identity $\mathrm{adj}(M)_{ii} = \det(M\!\restriction_{j\neq i})$ over an arbitrary finite index type, which we prove by a block-triangular reindexing $n \simeq \{i\} \sqcup \{j \neq i\}$ rather than the `Fin`-only cofactor expansion Mathlib carries (`adjugate_diag_eq_det_submatrix_ne`).

From the submatrix to the isolated root. Part III's spectral form deletes the root as a matrix *index*; Part II's divisibility identity speaks of deleting its *edges*. The two must be reconciled. Deleting all edges at the root r of $T = T(G, v)$ leaves a zero row, so $1 - X\,A(T_{\widehat{r}})$ has its r-th row equal to the basis vector e_r, and its determinant collapses to the $\{j \neq r\}$ minor—which is exactly the submatrix the resolvent produced. Hence

$$\boxed{\overline{\mathrm{charpolyRev}(A(T_{\widehat{r}}))} \;=\; \overline{\mathrm{charpolyRev}(A(T)\!\restriction_{j\neq r})},}$$

the bridge between the two deletions (`coe_charpolyRev_adjMatrix_deleteIncidenceSet`, resting on the general `det_eq_det_submatrix_ne_of_row_eq_single`).

The per-vertex identity. Now Part II enters. Reversing the divisibility identity $\mu(G)\,\mu(T_{\widehat{r}}) = \mu(G - v)\,\mu(T)$ over the integral domain $\mathbb{R}[X]$ (reversal is multiplicative there, and $\mathrm{rev}\circ\det = \mathrm{charpolyRev}$), the forest bridge $\mu(T) = \det(xI - A(T))$ turns each reversed matching polynomial into a charpolyRev; cancelling the common unit factor $\overline{\mathrm{charpolyRev}(A(T))}$ collapses the whole per-tree story to a single clean statement about G:

$$\boxed{\Big(\sum_{k}[A(T(G,v))^k]_{\mathrm{root}}X^k\Big)\cdot\overline{\mu(G)} \;=\; \overline{\mu(G-v)}}$$

(`resolventGenfun_pathTree_mul_reverse_matchingPoly`; here $G - v$ is the incidence-deletion form, isolating v). The path tree has done its work and vanished: only $\mu(G)$ and $\mu(G - v)$ remain.

Summing over v. Summing the per-vertex identity over all vertices, the left side becomes $W(G)\cdot\overline{\mu(G)}$ (Part III's spectral form, with an $\mathbb{N} \to \mathbb{R}$ cast supplied by `Matrix.map_pow`). The right side is $\sum_v \overline{\mu(G - v)}$, and here the reflection bookkeeping pays off: every $\mu(G-v)$ has the same degree n, so the reversals share a length and $\sum_v \mathrm{rev}(\mu(G - v)) = \mathrm{reflect}_n(\sum_v \mu(G - v))$ (`sum_reverse_eq_reflect_sum`). The vertex-deletion derivative law $\sum_v \mu(G - v) = X\,\mu'(G)$ (`sum_matchingPoly_deleteIncidenceSet`, proved in the companion to Part II) finishes the move:

$$\boxed{W(G) \cdot \overline{\mu(G)} \;=\; \overline{X\,\mu'(G)}.}$$

The entire trace side is now one reflected matching polynomial (`mk_treeLikeWalkCount_mul_reverse_eq`). Using $\mathrm{reflect}_n$, rather than rev, for the numerator is what lets us avoid ever computing the degree of $X\mu'$.

4 The matching side becomes the same

The matching side knows nothing of trees; it knows only the roots. How does the generating function of $\sum_i \theta_i^k$ become the very same $\overline{X\mu'}$? Through a geometric series that cancels a product, and the classical fact that a polynomial's derivative is the sum of its leave-one-out factor products.

Power sums as geometric series. The generating function of the power sums is, coefficient by coefficient, the multiset-sum of per-root geometric series,

$$M(G) \;=\; \sum_{\theta} \mathrm{geomSeries}(\theta), \qquad \mathrm{geomSeries}(\theta) = \sum_k \theta^k X^k = (1-\theta X)^{-1}$$

(`powerSum_genfun`). Multiplying by the reversed factor product collapses it, root by root: since $\mathrm{geomSeries}(\theta)\,(1-\theta X) = 1$ and the whole product $\prod_\theta (1-\theta X)$ factors as $(1-\theta X)$ times the leave-one-out product, one obtains

$$\boxed{\Big(\sum_{\theta} \mathrm{geomSeries}(\theta)\Big) \cdot \prod_{\theta}(1-\theta X) \;=\; \sum_{\theta}\prod_{\varphi\neq\theta}(1-\varphi X)}$$

with *no* derivative and *no* Newton recursion—only the cancellation of one factor (`geomSeries_sum_mul_prod`). Because $\mu_{\mathbb{C}}$ is monic and splits over $\mathbb{C}$, it equals $\prod_\theta (X-\theta)$, and reversal sends this to $\prod_\theta (1-\theta X)$ (`reverse_prod_X_sub_C`); so the left factor is $\overline{\mu_{\mathbb{C}}}$.

The leave-one-out sum is the reflected derivative. The right-hand side is the matching mirror of the trace side's numerator. Classically, for a split polynomial, $\mu'_{\mathbb{C}} = \sum_\theta \prod_{\varphi\neq\theta}(X-\varphi)$ (`derivative_prod_X_sub_C`, the product rule). Each leave-one-out product has degree $n-1$, so $X \cdot \prod_{\varphi\neq\theta}(X-\varphi)$ has degree n; reflecting at the fixed length n and using $\mathrm{rev}(Xq) = \mathrm{rev}(q)$ termwise,

$$\boxed{\sum_{\theta}\prod_{\varphi\neq\theta}(1-\varphi X) \;=\; \mathrm{reflect}_n(X\,\mu'_{\mathbb{C}})}$$

(`reflect_X_mul_derivative_eq_sum_prod`). Chaining the two boxes through the power-series coercion of a product and a sum gives the matching analogue of the trace identity,

$$\boxed{M(G) \cdot \overline{\mu_{\mathbb{C}}} \;=\; \overline{X\,\mu'_{\mathbb{C}}}}$$

(`mk_powerSum_mul_reverse`, specialized to $\mu_{\mathbb{C}}$ in `mk_matchingPowerSum_mul_reverse_eq`). Both sides now speak the same reflected polynomial.

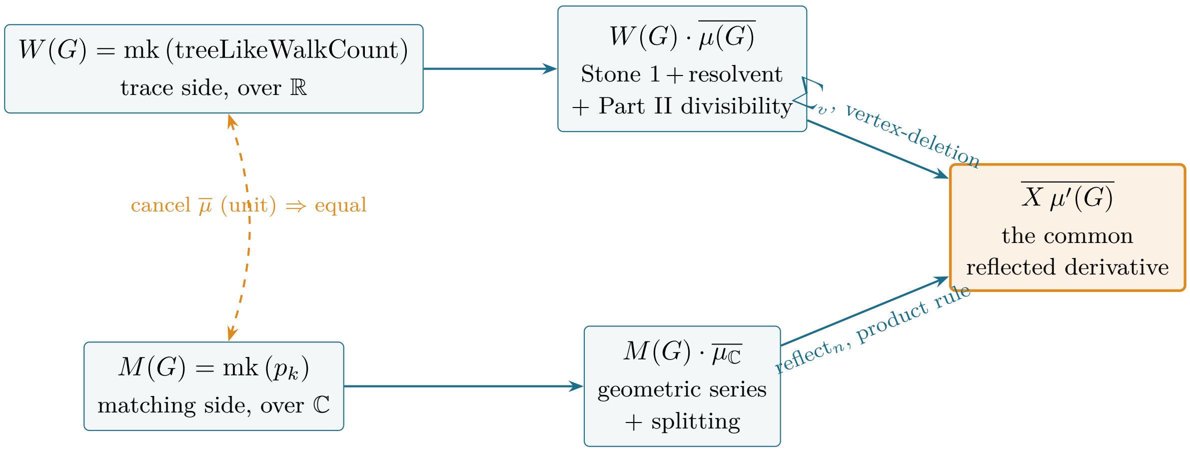

Figure 2: The weld. Both generating functions, multiplied by the reversed matching polynomial, are forced to equal the same reflected derivative $\overline{X\mu'(G)}$ (over $\mathbb{R}$ on top, over $\mathbb{C}$ below, joined by the base change $\mathbb{R} \to \mathbb{C}$). Cancelling the unit $\overline{\mu}$ identifies the two series; the coefficient of X^k is the moment theorem.

5 The weld

Two identities, one over $\mathbb{R}$ and one over $\mathbb{C}$, both ending at $\overline{X\mu'}$. What is left but to put them together? A change of base and a cancellation.

The trace identity lives over $\mathbb{R}$; the matching one over $\mathbb{C}$. Mapping the former along the ring homomorphism $\mathbb{R} \to \mathbb{C}$ commutes with everything in sight—with mk (through `coeff_map` and $\mathbb{N}$-casts), with the polynomial-to-series coercion, with rev (which is reflect at a degree preserved by an injective map), with reflect, and with the derivative (`reflect_map`, `derivative_map`, `natDegree_map_eq_of_injective`). The image of the trace identity is therefore

$$W_{\mathbb{C}}(G) \cdot \overline{\mu_{\mathbb{C}}} \;=\; \overline{X\,\mu'_{\mathbb{C}}}, \qquad W_{\mathbb{C}}(G) = \mathrm{mk}\big(k \mapsto (\mathrm{treeLikeWalkCount}(G,k) : \mathbb{C})\big),$$

identical in its right-hand side to the matching identity $M(G) \cdot \overline{\mu_{\mathbb{C}}} = \overline{X\mu'_{\mathbb{C}}}$. Hence $W_{\mathbb{C}}(G) \cdot \overline{\mu_{\mathbb{C}}} = M(G) \cdot \overline{\mu_{\mathbb{C}}}$. The reversed matching polynomial $\overline{\mu_{\mathbb{C}}}$ is a unit: $\mu_{\mathbb{C}}$ is monic, so $\mathrm{rev}(\mu_{\mathbb{C}})$ has constant term 1 and is nonzero, and $\mathbb{C}[\![X]\!]$ is an integral domain. Cancelling it gives $W_{\mathbb{C}}(G) = M(G)$, and reading the coefficient of X^k is the theorem.

Theorem 1 (Godsil's moment theorem, machine-checked)**.** *For every finite graph G on the vertex type V, every $k \in \mathbb{N}$,*

$$p_k(G) \;=\; \sum_{\theta \in \mathrm{roots}(\mu_{\mathbb{C}})} \theta^k \;=\; (\mathrm{treeLikeWalkCount}(G,k) : \mathbb{C}).$$

In Lean: `matchingPowerSum_eq_treeLikeWalkCount`. The proof is `sorry`-free and depends only on `propext`, `Classical.choice`, `Quot.sound`.

6 Numerical verification

A formal proof certifies the implication; what certifies that the statement formalized is the one intended? Every identity in this paper was de-risked numerically before any formal effort, against exact arithmetic, and the same checks fix the theorem's two sides side by side. It is worth seeing the theorem land on one small graph in full.

Table 2: Matching power sums $p_k = \sum_i \theta_i^k$, computed from the roots of $\mu(G)$ in exact $\overline{\mathbb{Q}}$ arithmetic, equal the path-tree walk count treeLikeWalkCount(G, k) of Part III for every G and k (the content of Theorem 1). Odd p_k vanish. The final column is the first trace-formula gap $\mathrm{tr}(A^g) - p_g$ at $k =$ girth g, matching the shortest-cycle law $2g\,c_g$.

G	n	girth	p_0	p_2	p_4	p_6	$\mathrm{tr}(A^g) - p_g = 2g\,c_g$
K_4	4	3	4	12	60	324	$24 = 2 \cdot 3 \cdot 4$
C_5	5	5	5	10	30	100	$10 = 2 \cdot 5 \cdot 1$
$K_{3,3}$	6	4	6	18	90	522	$72 = 2 \cdot 4 \cdot 9$
Petersen	10	5	10	30	150	870	$120 = 2 \cdot 5 \cdot 12$

A worked example: the pentagon C_5. Take the 5-cycle. Its matchings are the empty one, the five single edges, and the five pairs of non-adjacent edges, so $m_0 = 1$, $m_1 = 5$, $m_2 = 5$ and

$$\mu(C_5, x) \;=\; x^5 - 5x^3 + 5x \;=\; x\Big(x^2 - \tfrac{5+\sqrt{5}}{2}\Big)\Big(x^2 - \tfrac{5-\sqrt{5}}{2}\Big),$$

with roots $\theta \in \{0,\ \pm\sqrt{(5+\sqrt{5})/2},\ \pm\sqrt{(5-\sqrt{5})/2}\}$. The *matching* side reads the power sums off these roots:

$$p_2 = \sum_i \theta_i^2 = \tfrac{5+\sqrt{5}}{2} \cdot 2 + \tfrac{5-\sqrt{5}}{2} \cdot 2 = 10, \qquad p_4 = \sum_i \theta_i^4 = \Big(\tfrac{5+\sqrt{5}}{2}\Big)^2 \cdot 2 + \Big(\tfrac{5-\sqrt{5}}{2}\Big)^2 \cdot 2 = 30.$$

The *tree-like* side counts walks. For $k = 2$, a closed tree-like walk at a vertex steps to a neighbour and back; each of the 5 vertices has 2 neighbours, giving $2 \cdot 5 = 10 = p_2$. For $k = 4$, the girth of C_5 is 5, so *every* closed walk of length 4 is tree-like (none can yet enclose the pentagon), and the count is $\mathrm{tr}(A^4) \;=\; 2^4 + 2\big(\tfrac{\sqrt{5}-1}{2}\big)^4 + 2\big(\tfrac{\sqrt{5}+1}{2}\big)^4 \;=\; 30 \;=\; p_4$, using the adjacency eigenvalues $2,\ \tfrac{\sqrt{5}-1}{2}\,(\times 2),\ -\tfrac{\sqrt{5}+1}{2}\,(\times 2)$. The two sides agree, as Theorem 1 requires. The first *disagreement* is the theorem's other content: at $k = 5 =$ girth, $p_5 = 0$ (the polynomial is odd) while $\mathrm{tr}(A^5) = 10$, so the gap is $10 = 2 \cdot 5 \cdot 1$—the single pentagon, traversed from 5 basepoints in 2 orientations. The 5-cycle is exactly the walk that is closed but not tree-like, and it appears, on the nose, at length 5.

Figure 3 plots the two sides of the trace formula on K_4 and the Petersen graph: the tree side $p_k = \sum_i \theta_i^k =$ treeLikeWalkCount(G, k) (the content of Theorem 1) against the cycle side $\mathrm{tr}(A^k)$, the count of *all* closed walks. Below the girth every closed walk is tree-like, so the curves coincide; the first gap appears exactly at $k =$ girth and equals $2g\,c_g$, twice the girth times the number of shortest cycles—24 on K_4 ($g = 3$, $c_3 = 4$), 120 on Petersen ($g = 5$, $c_5 = 12$). Table 2 records the matching power sums, computed from the roots of the matching polynomial, on a wider sample; in each case they agree with the path-tree walk count of Part III and with the girth-gap law. The matching polynomials themselves were checked in exact $\overline{\mathbb{Q}}$ arithmetic (`SageMath`); the underlying generating-function identities (the geometric-series cancellation ($\star$), the reversed log-derivative $\sum_k p_k z^k = n - z\,q'/q$, and the reverse-algebra of the assembly) were each confirmed to exact equality before the corresponding Lean lemma was written.

7 The boundary, now closed—and the route the prequel did not predict

Part III mapped this completion as "the single-entry resolvent identity and the univariate Newton step." The resolvent step is here as foreseen. The Newton step is not—and its absence is the one mathematically interesting thing about the proof.

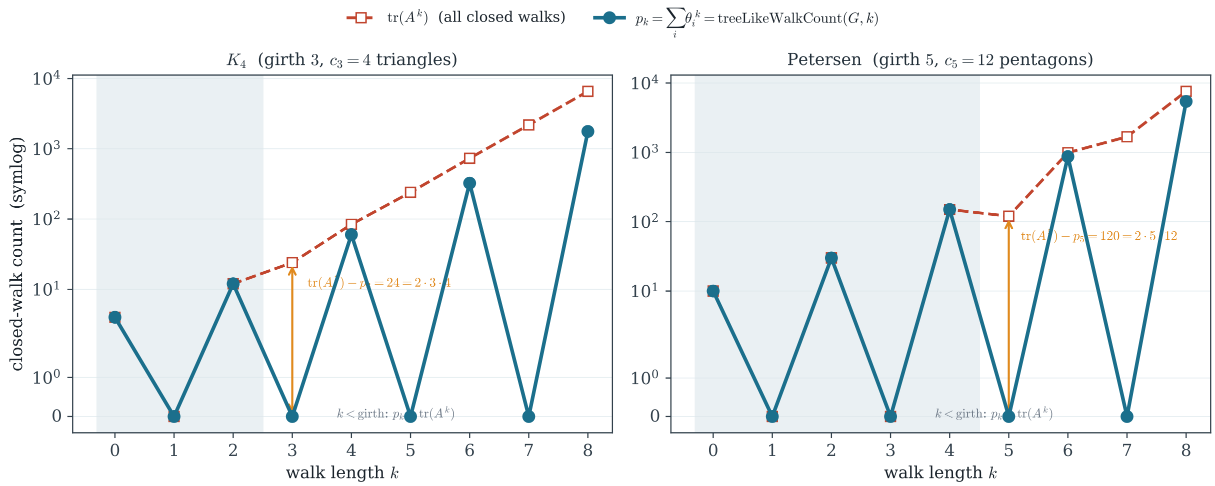

Figure 3: The moment theorem as the tree side of the trace formula. For K_4 (girth 3) and the Petersen graph (girth 5), the matching power sums $p_k = \sum_i \theta_i^k$ (teal), which Theorem 1 certifies equal to the tree-like walk count $\text{treeLikeWalkCount}(G, k) = \sum_v [A(T(G, v))^k]_{\text{root}}$, against $\text{tr}(A^k)$ (red), the number of all closed walks. The two coincide below the girth (shaded); the first gap, $\text{tr}(A^g) - p_g = 2g\, c_g$, counts the shortest cycles (g basepoints $\times\, 2$ orientations $\times\, c_g$ cycles). Odd power sums vanish because the matching polynomial is even. The cycle side is the *Bass* companion of this series; the tree side is this paper.

Part III's boundary section sketched a five-link chain to the moment theorem: the resolvent identity, the principal-minor lemma, the forest bridge, the divisibility identity, the vertex-deletion law, and then a *univariate Newton identity* equating the logarithmic derivative μ'/μ with $\sum_k p_k X^k$. Four of those links are exactly what Sections 3–5 use. The fifth is not. There is no Newton recursion in the formalization, and no univariate-root Newton lemma was added to Mathlib's staging.

What replaced it is the cancellation of Section 4: the generating function $\sum_\theta (1 - \theta X)^{-1}$ multiplied by $\prod_\theta (1 - \theta X)$ telescopes, by one factor's worth of cancellation, into the leave-one-out sum $\sum_\theta \prod_{\varphi \neq \theta} (1 - \varphi X)$, which the product rule and a reflection identify with $\overline{X\mu'}$. Newton's identities relate power sums to elementary symmetric functions through a recursion; here the same content appears as a single multiset cancellation, with the derivative supplied by the product rule rather than by a recursion on p_k. The logarithmic-derivative reading $\sum_k p_k X^k = n - X\, q'/q$ (with $q = \text{rev}(\mu)$) is still true—it was the numerical anchor that de-risked the step—but the formal proof never forms q'/q; it works with cleared denominators throughout, which is why it needs no inverse and no recursion. This is the kind of small surprise a formalization is good at surfacing: the route that looks canonical on paper (Newton) is not the cheapest route past the machine.

The principal-minor lemma Part III flagged as "the one genuinely new linear-algebra lemma" is present (`adjugate_diag_eq_det_submatrix_ne`), and so is its determinant sibling for an isolated row (`det_eq_det_submatrix_ne_of_row_eq_single`); both are graph-free and kept in canonical form. The reflection-of-a-sum lemma (`sum_reverse_eq_reflect_sum`) and the reverse of a split product (`reverse_prod_X_sub_C`) are likewise general. Together with the cycle side in the *Bass* companion [18], the finite matching/Ihara trace formula now has both of its ends—$\text{tr}(A^k)$ counting all closed walks and p_k counting the tree-like ones—on certified ground in one library; below the girth their coefficients coincide as theorems, and the first gap is a count of shortest cycles, waiting to be read off.

8 Implications and methodology

The formalization and manuscript were developed with AI assistance (Claude, Anthropic); the author set the research targets, judged the mathematics, and is responsible for all claims. Every theorem named here compiles `sorry`-free against the pinned Mathlib revision [20] and depends only on the three standard axioms, re-verifiable with `#print axioms`.

A false green, and the lesson banked. One identity (`godsil_resolvent_charpoly_form`) was, for a time, silently broken: an incremental `lake build` replayed a stale compiled object and reported success while the source no longer compiled. The lesson is worth banking—a passing per-target build is not proof that a theorem compiles; only a clean elaboration of the file is—and every identity in this paper was re-checked by elaborating its module from source.

Where it applies, and how. The moment theorem is the bridge between two ways of reading a graph: as a combinatorial object whose walks one counts, and as the carrier of a spectrum—a ghost one, not the graph's adjacency spectrum—whose moments one bounds. It is precisely the identity invoked, usually tacitly, wherever one of these is estimated through the other. Concretely:

- *Eigenvalue bounds by the method of moments.* The largest matching root obeys $\theta_{\max} = \lim_k p_{2k}^{1/2k}$, and the moment theorem makes each p_{2k} a *walk count* on the path tree—an explicitly bounded combinatorial quantity, dominated by walks on the infinite $(\Delta-1)$-ary universal cover, whose spectral density is the Kesten–McKay law [11, 9]. This is the route to Godsil's bound $\theta_{\max} \le 2\sqrt{\Delta-1}$, the matching analogue of the spectral-radius estimate, and the certified p_k is its raw material.
- *Expander and Ramanujan thresholds.* The band edge $2\sqrt{\Delta-1}$ separating a Δ-regular graph's trivial eigenvalue from the rest is the same number; a Ramanujan graph is one whose nontrivial spectrum stays inside it [15]. The moment theorem ties the matching-root version of that threshold to a tree-like walk count, the form in which the universal-cover comparison is cleanest.
- *Statistical mechanics.* $\mu(G,x)$ is the monomer–dimer partition function; its roots are the zeros that control the absence of a phase transition (Heilmann–Lieb [1]), and their power sums p_k are the cumulant-like moments a high-temperature expansion produces. The theorem reads those moments off the graph's tree-like walks.
- *Networks via the non-backtracking operator.* The cycle side of the trace formula, $\mathrm{tr}(A^k)$ and its non-backtracking refinement $\mathrm{tr}(B^k)$, governs percolation and community-detection thresholds on networks [13, 14]; the moment theorem supplies the tree side against which those cycle counts are measured, and the first gap, $\mathrm{tr}(A^g) - p_g = 2g\,c_g$, is a certified shortest-cycle count.

This paper adds no new bound and no algorithm to any of these. It places the *bridge itself*—the exact identity each of them crosses—on certified ground, in the same library where the path tree already carries a real spectrum (Part II) and counts tree-like walks (Part III). The same forest now does a third job: it makes the count *equal* the spectrum. A foundation, and the closing of a four-part arc.

The formalized scaffold. Table 3 lists the load-bearing results, their Lean names, and where each is proved; the general, graph-free lemmas (marked $*$) are kept in canonical form for upstreaming to Mathlib.

Table 3: The load-bearing formalized results. All are `sorry`-free against the pinned Mathlib revision; $*$ marks the general, graph-free lemmas absent from Mathlib and kept for upstreaming.

Lean name	Statement	§
$*$`det_eq_det_submatrix_ne_of_row_eq_single`	row $i = e_i \Rightarrow \det M = \det(M\restriction_{\neq i})$	3
`coe_charpolyRev_adjMatrix_deleteIncidenceSet`	$\overline{\text{charpolyRev}\, A(G_{\widehat{r}})} = \overline{\text{charpolyRev}(A(G)\restriction_{\neq r})}$	3
`resolventGenfun_diag_mul_coe_charpolyRev`	$\left(\sum_k [M^k]_{ii} X^k\right)\overline{\text{charpolyRev}\, M} = \overline{\text{charpolyRev}(M\restriction_{\neq i})}$	3
`resolventGenfun_pathTree_mul_reverse_matchingPoly`	per-vertex: $\left(\sum_k [A(T)^k]_{\text{root}} X^k\right)\overline{\mu(G)} = \overline{\mu(G-v)}$	3
$*$`sum_reverse_eq_reflect_sum`	$\sum_i \text{rev}(p_i) = \text{reflect}_N(\sum_i p_i)$ if $\deg p_i = N$	3
`mk_treeLikeWalkCount_mul_reverse_eq`	trace side: $W(G)\,\overline{\mu(G)} = \overline{X\mu'(G)}$	3
`geomSeries_sum_mul_prod`	$\left(\sum_\theta \frac{1}{1-\theta X}\right)\prod_\theta(1-\theta X) = \sum_\theta \prod_{\varphi\neq\theta}(1-\varphi X)$	4
$*$`reverse_prod_X_sub_C`	$\text{rev}\prod_\theta(X-\theta) = \prod_\theta(1-\theta X)$	4
`reflect_X_mul_derivative_eq_sum_prod`	$\sum_\theta \prod_{\varphi\neq\theta}(1-\varphi X) = \text{reflect}_n(X\mu')$	4
`mk_matchingPowerSum_mul_reverse_eq`	matching side: $M(G)\,\overline{\mu_{\mathbb{C}}} = \overline{X\mu'_{\mathbb{C}}}$	4
`matchingPowerSum_eq_treeLikeWalkCount`	**moment theorem:** $p_k(G) = \text{treeLikeWalkCount}(G,k)$	5

9 Questions left open

A formalization is also a list of the next questions, sharpened until a machine could be asked them.

- The proof routes around Newton's identity. Is the geometric-series cancellation *equivalent* to the univariate Newton recursion as a Mathlib lemma—i.e. does one give a short proof of the other—or are they genuinely different normal forms of the same content, one recursive and one closed?
- Both sides were forced to equal $\overline{X\mu'}$. Can the proof be refactored so that the two generating functions are identified *before* the reflection bookkeeping, purely as ratios of reversed polynomials, leaving reflect_n out of the cancellation entirely?
- Below the girth the moment theorem gives $p_k = \text{tr}(A^k)$, and the first gap at $k = \text{girth}$ counts shortest cycles. Now that both sides are formal objects, can the gap $\text{tr}(A^k) - p_k$ be evaluated at $k = \text{girth}$ inside the library, yielding the cycle-counting correction $2\,\text{girth}\cdot c_g$ as a theorem?
- The reversed-polynomial bookkeeping (rev, reflect_N, the unit $\overline{\mu}$) recurs in both the matching and the Ihara/Bass sides. Is there a single *reversed generating-function* calculus that subsumes both, so that the full trace formula $\text{tr}(A^k) = p_k + (\text{cycle terms})$ is one identity rather than two welded halves?
- The principal-minor and reflection lemmas are general. Pushed upstream, would they shorten existing Mathlib developments—resultants, companion matrices, the existing `charpolyRev` API—or are they only ever useful in the company of a path tree?

And one to keep. *The matching polynomial is not the characteristic polynomial of the graph's own adjacency matrix, yet its root power sums count walks exactly as a spectrum's would; the theorem* *says where those moments live—but is the spectrum they describe a bookkeeping artefact of the path tree, or, for every purpose that consumes its moments, as real as any matrix's?*

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